ELEKTRONIKA ANALOG

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1.2 Ideal Diode

2). Describe in your own words the characteristics of the ideal diode and how they determine the on and OFF states of the device. That is, describe why the short circuit and open circuit equivalents are appropriate.

↳ karakteristik dioda sangat mirip dengan sakelar sederhana yang dapat menghantarkan arus dalam satu arah yang ditentukan oleh panah pada simbol yaitu dari anoda ke katoda. pada bias maju, tegangan positif akan melintasi dioda ~~akan~~ dan menghasilkan konduksi keadaan "on", pada bias mundur, tegangan balik melintasi dioda dengan resistansi yang tinggi maka konduksi keadaan "OFF" atau rangkaian terbuka.

1.3 Semiconductor Materials

4). In your own words, define semiconductor, resistivity, bulk resistance, and ohmic contact resistance

↳ • Semikonduktor = bahan dengan karakteristik konduktivitas yang terletak diantara isolar dan konduktor

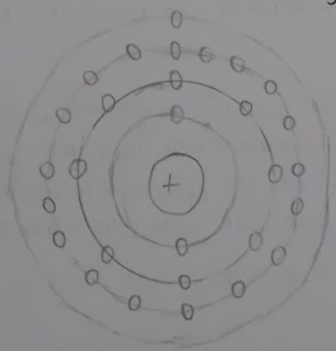
• Resistivitas = karakteristik bahan yang akan menentukan tingkat perlawanan terhadap aliran arus melalui beban

• Resistansi massal = resistansi dioda ketika dalam bias maju dalam pusaran beberapa ohm, nilainya tergantung ukuran bahan p dan n

• Resistansi kontak ohmic = resistansi yang ditimbulkan oleh hubungan antara tanah logam dan bahan semikonduktor

6). Sketch the atomic structure of copper and discuss why it is a good conductor and how its structure is different from germanium and silicon

↳



Tembaga memiliki 29 elekteron yang mengorbit dengan 1 elekteron dikulit terluar karena subkulit dapat berisi 2 elektron maka kulit terluar tidak lengkap dan jauh dari nukleus menunjukkan bahwa elektron terikat secara longgar pada atom induknya. Germanium dan silikon memiliki kulit terluar yang lengkap karena berbagi (ikatan kovalen) elektron antara atom. Elektron yang merupakan bagian dari struktur kulit yang lengkap memerlukan peningkatan tingkat gaya tarik menarik untuk dikeluarkan dari atom induknya.

8). Consult your referency library and list three materials that have a negative temperature coefficient and three that have a positive temperature coefficient

↳ koefisien suhu negatif yaitu bahan semikonduktor = karbon, silikon, germanium
koefisien suhu positif yaitu logam murni = tembaga, perak, alumunium

1.4 Energy Levels

10). If .48 eV of energy is required to move a change through a potential difference of 12 V, determine the charge involved.

↳ $48\ eV = 48\ (1{,}6 \times 10^{-19}) = 76{,}8 \times 10^{-19}\ J$

$$Q = \frac{w}{V} = \frac{76{,}8 \times 10^{-19}\ J}{12\ V} = 6{,}40 \times 10^{-19}\ C$$

1.5 Extrinsic Materials - n and p type

12). Describe the difference between n type and p type semiconductor materials

↳ Bahan semikonduktor tipe n memiliki kelebihan elektron untuk konduksi yang dibuat dengan mendoping bahan intrinsik dengan atom donor yang memiliki elektron valensi lebih banyak daripada yang diperlukan untuk membuat ikatan kovalen. ~~sehingga menciptakan hole pada~~ pembawa mayoritas adalah elektron, pembawa minoritas adalah hole.

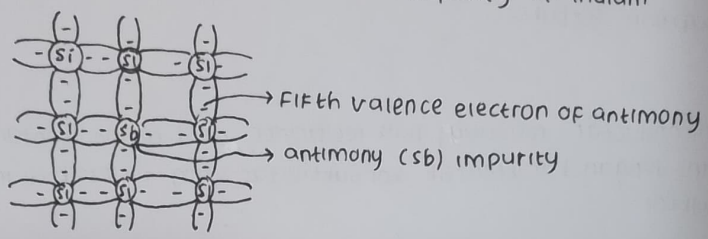
Bahan semikonduktor tipe P dibentuk dengan mendoping bahan intrinsik dengan atom akseptor yang memiliki jumlah elektron yang tidak mencukupi pada kulit valensi untuk melengkapi ikatan kovalen sehingga menciptakan hole pada struktur buatan kovalen. pembawa mayoritas adalah hole dan minoritas adalah elektron

14). Describe the difference between majority and minority carriers

↳ * pembawa mayoritas = pembawa bahan yang jauh melebihi jumlah pembawa lain dalam bahan
* pembawa minoritas = pembawa material yang jumlahnya lebih sedikit daripada pembawa material lain

16). Repeat problem 15 but insert an impurity of indium

↳



1.6 Semiconductor Diode

18). Describe in your own conditions established by forward and reverse bias conditions on a p-n junction diode and how the resulting current is affected

↳ * kondisi bias maju : terminal positif baterai terhubung ke sisi p dan terminal negatif ke n tegangan yang maju menciptakan medan listrik yang bekerja melawan medan karena penghalang potensial. ini melemahkan medan yang dihasilkan sehingga ketinggian penghalang berkurang. penghalang potensial yang berkurang ini memungkinkan aliran pembawa muatan melintasi persimpangan yang mengarah ke resistansi rendah dan dengan demikian arus maju

* kondisi bias mundur : - baterai terhubung ke p dan + ke n. tegangan balik yang diterapkan menciptakan medan listrik yg bekerja dalam arah yg sama dengan medan penghalang potensial yg meningkat ini mencegah aliran pembawa muatan melintasi persimpangan yg mengarah ke resistansi tinggi dan dgn demikian tidak ada

20). using Eq (1.4) determine the diode current at 20°C for a silikon diode with I_s = 50 nA and an applied forward bias of 0.6 V

↳ $T_k = 20 + 273 = 293$

$R = \frac{11.600}{n} = \frac{11.600}{2} = 5800$

$I_D = I_s (e^{\frac{kV_D}{T_k}} - 1)$

$= (e^{\frac{(5800)(0.6)}{293}} - 1)$

$= 50 \times 10^{-9} (e^{11877} - 1) = 7.197$ mA

22). a). using Eq (1.4) determine the diode current at 20°C for a silikon drode with I_s = 0.1 μA at a reserve bias potential of -10 V

↳ $20 + 273 = 293$

$R = \frac{11.600}{n} = \frac{11.600}{2} = 5800$

$I_D = I_s (e^{\frac{kV_D}{T_k}} - 1)$

$= 0.1\ \mu A\ (e^{\frac{(5800)(-10V)}{293}} - 1) = 0.1 \mu A\ (e^{-197.93} - 1)$

$= 0.1 \times 10^{-6}\ (1.07 \times 10^{-86} - 1) \cong 0.1 \times 10^{-6} = 0.1\ \mu A$

b). Is the result expected? why?

↳ hasilnya diharapkan dan arus dalam kondisi bias balik harus sama dengan nilai satu rapi

24) In the reserve bias region the saturation current of a silicon diode is about 0,1 μA ($T = 20°C$). Determine its approximate value if the temperature is increased 40°C

↳ $T = 20°C : I_S = 0,1\,\mu A$ $\quad T = 50°C : I_S = 2(0,4\,\mu A) = 0,8\,\mu A$

$T = 30°C : I_S = 2(0,1\,\mu A) = 0,2\,\mu A$ $\quad T = 60°C : I_S = 2(0,8\,\mu A) = 1,6\,\mu A$

$T = 40°C : I_S = 2(0,2\,\mu A) = 0,4\,\mu A$ $\quad 1,6\,\mu A = 0,1\,\mu A \rightarrow 16 : 1$ meningkat karena kenaikan suhu 40°C

26). Determine the forward voltage drop across the diode whose characteristics appear in Fig 1.24 at temperatures of -75°C, 25°C, 100°C and 200°C and a current of 10 mA. For each temperature, determine the level of saturation current. compare the extremes of each and comment on the ratio of the two

↳ $-75°C = V_F = 1,7\,V$ $\quad I_S = 0,1\,\mu A$

$25°C = V_F = 1,3\,V$ $\quad I_S = 0,5\,\mu A$

$100°C = V_F = 1,0\,V$ $\quad I_S = 1\,\mu A$

$200°C = V_F = 0,65\,V$ $\quad I_S = 2\,\mu A$

V_F menurun dengan meningkatkan suhu

$1,7V : 0,65\,V \approx 2,6 : 1$

I_S meningkat dengan meningkatkan suhu

$2\,\mu A : 0,1\,\mu A = 20 : 1$

1.7 Resistance levels

28). Repeat problem 26 at a forward current of 15 mA and compare results

↳ $I_D = 5\,mA$, $V_D = 0,82\,V$ $\quad R_{DC} = \frac{V_D}{I_D} = \frac{0,82\,V}{5\,mA} = 54,67\,\Omega$

30). a). Determine the dynamic (ac) resistance of the diode of Fig 1.29 at a forward current of 10 mA using Eq (1.6)

↳ $r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{0,79\,V - 0,76\,V}{15\,mA - 5\,mA} = \frac{0,03\,V}{10\,mA} = 3\,\Omega$

b). Determine the dynamic (ac) resistance of the diode of Fig 1.29 at a forward current of 10 mA using Eq (1.7)

↳ $r_d = \frac{26\,mV}{I_D} = \frac{26\,mV}{10\,mA} = 2,6\,\Omega$

c). compare solutions of parts (a) and (b)

↳ cukup dekat perbedaannya

32). using Eq (1.7) determine the ac resistance at a current of 1mA and 15mA for the diode of Fig 1.29 compere the solution and develop a general conclusion regarding the ac resistance and increasing levels of diode current

↳ $I_D = 1\,mA = r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{0,72\,V - 0,61\,V}{2\,mA - 0\,mA} = 55\,\Omega$

$I_D = 15\,mA = r_d = \frac{\Delta V_d}{\Delta I_d} = \frac{0,8\,V - 0,78\,V}{20\,mA - 10\,mA} = 2\,\Omega$

34). Determine the avarege ac resistance for the diode of Fig 1.19 for the region between 0.6 and 0.9 V

↳ $r_{av} = \frac{\Delta V_d}{\Delta I_d} = \frac{0,9\,V - 0,6\,V}{13,9\,mA - 12\,mA} = 24,4\,\Omega$

1.8 Diode Equivalent circuits

36). Find the piecewise linear equivalent circuit for the diode of Fig 1.19 use a straight line segment that intersects the horizontal axis at 0.7 V and best approximates the curve for the region

↳ $r_d = \frac{nV_T}{I_D}$ = asumsikan $n = 1$ $\quad \frac{25\,mV}{10\,mA} = 2,5\,\Omega$

38). Plot I_F versus V_F using linear scales for the diode of Fig 1.36 Note that the provided graph employs as log scale for the vertical axis (log scales are covered in sections 11.2 11.3)

↳

40). Does the reverse saturation current of the diode of Fig 1.36 change significantly in magnitude for reverse bias potentials in the range -25 to -100 V ?

↳ arus reverse tidak berubah secara signifikan dalam rentang -25 V hingga -100 V

42). For the diode of Fig 1.36 determine the maximum ac (dynamic) resistance at a forward current of 0.1, 1.5 and 20mA. Compare levels and comment on whether the results support conclusions derived in earlier sections of this chapter

↳ $V_T = \frac{kT}{q} \approx 25mV$

$r_d = \frac{nV_T}{I_F}$

$I_F = 0.1\,mA = 0.0001\,A$

$r_d = \frac{1.25mV}{0,0001A} = \frac{0.025V}{0,0001A} = 250\,\Omega$

$I_F = 1.5\,mA = 0,0015\,A$

$r_d = \frac{1.25mV}{0,0015} = \frac{0,025V}{0,0015A} = 16.67\,\Omega$

$I_F = 20mA = 0,02\,A$

$r_d = \frac{1.25mV}{0,02\,A} = \frac{0,025V}{0,02\,A} = 1.25\,\Omega$

44). using the characteristics of Fig. 1.36, determine the temperature at which the diode current will be 50% of its value at room temperature (25°C)

↳ $\frac{I_F(T)}{I_F(25°C)} = \frac{I_S(T)}{I_S(25°C)}$

Jika 50% diturunkan = $0,5 = e^{\frac{(T-25)}{k}}$

$T-25 = k\ln(0,5)$

$T-25 = 10.\ln(0,5)$

$T-25 = 10.(-0,693)$

$T-25 = -6.93$

$T = 18°C$

✱ 1.10 Transition and diffusion capacitance

46). Referring to Fig 1.31 determine the diffusion capacitances ~~differ~~ at 0 and 0.25 V

↳

48). determine the reactance offered by a diode described by the characteristics of Fig 1.37 at a forward potential of 0,2V and a reserve potential of -20 V if the applied frequency is 6 MH_2

↳ $X = \frac{1}{2\pi FC}$ $\quad C = 10pF = 10 \times 10^{-12} F$

$X = \frac{1}{2\pi \times 6 \times 10^6 \times 10 \times 10^{-12}} = 2.65 \Omega$

tegangan maju (0,2 V)

tegangan balik

$X = \frac{1}{2\pi \times 6 \times 10^6 \times 1 \times 10^{-12}}$

$= 26.5 \Omega$

* 1.14 zener diodes

50). The following characteristics are specified for a particular zener diode $V_z = 29$ V $V_R = 16.8$ V, $I_{zr} = 10mA$, $I_R = 20\mu A$ and $I_{zM} = 40mA$. Sketch the characteristics curve in the manner displayed in Fig 1.50

↳

52). Determine the temperature coefficient of a 5-V zener diode (rated 25°C value) if the nominal voltage drops to 4.8 V at a temperature of 100°C

↳ $T_1 = 25°C = 5V$

$T_2 = 100°C = 4.8V$

$\alpha = \frac{V_{T_2} - V_{T_1}}{V_{T_1} \times (T_2 - T_1)} = \frac{4.8V - 5V}{5V \times (100°C - 25°C)} = \frac{-0,2V}{5V \times 75°C} = \frac{-0.2V}{375V°C} = -0,0005333 V/$

~~54). Determine the impedance for the 24-V diode of Fig 1.51b at current levels of 0.2, 1 and 10mA. How do the results relate to the shape of the characteristics in this region?~~

54). Determine the impedance for the 24-V diode at $I_z = 10mA$ for Fig 1.51b. Note that it is a log sclale

↳ $z_d = \frac{V_z}{I_z} = \frac{24V}{10mA} = \frac{24V}{0,01A} = 2400 \Omega$

* 1.15 Light-Emitting Diodes

56). Reffering to Fig 1.55e what would appear to be an appropriate value of V_T for this device? How does it compare to the value of V_T for silicon & germanium?

↳ $V_T = \frac{kT}{q}$

$V_T = \frac{(1.38 \times 10^{-23} J \cdot 300k)}{1.6 \times 10^{-19} C} = 0,0259 V$

58). a). what is the percent increase in relative efficiency of the device of Fig 1.55 if the peak current is increased from 5 to 10 mA ?

↳ % increase = $\frac{100 - 25}{25} \times 100 = 300\%$

b). Repeat part a for 30 to 35 mA (the same increase in current)

↳ $(30\ mA)^2 = 900$

$(35\ mA^2) = 1225$

% increase = $\frac{1225 - 900}{900} \times 100 = 36.11\%$

c). compare the percent increase from parts (a) and (b). at what point on the curve would you say there is little gained by further increasing the peak current ?

↳ part a = 300% current from 5 to 10mA

part b = 36.11% dari 30 ke 35 mA

60). a). If the luminous intensity at 0° angular displacement is 3.0 mcd for the device of Fig 1.55 at what angle will it be 0.75 mcd ?

↳ $I(\theta) = I_0 \cos^n(\theta)$

$0.75 = 3.0 \cos^n(\theta)$

$\cos^n(\theta) = \frac{0.75}{3.0} = 0.25$

$\cos^2(\theta) = 0.25$

$\cos(\theta) = \sqrt{0,25} = 0.5$

$\theta = \cos^{-1}(0.5) = 60°$

b). Finding the angle where the luminous intensity drops below 50%

↳ $I(\theta) = I_0 \cos^n(\theta)$

$1.5 = 3.0 \cos^n(\theta)$

$\cos^n(\theta) = \frac{1.5}{3.0} = 0.5$

$\cos^2(\theta) = 0.5$

$\cos(\theta) = \sqrt{0.5} \approx 0.707$

$\theta = \cos^{-1}(0.707)$

$\theta \approx 45°$